

Lecture 4: Lifetimes

Practice smart pointers and lifetimes

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github.com/wvhulle/rust-course-ghent

0.1. Educational videos

For the people who are interested in dyn, dyn is an example of a dynamically-sized-type or unsized / ! Sized type. Read more about such types in <https://github.com/pretzelhammer/rust-blog/blob/master/posts/sizedness-in-rust.md>

Warning

If forgot to mention that dyn trait objects are a kind of type erasure. They erase size, alignment and other compile-time properties of the concrete types.

Youtubers that makes good videos about Rust:

- Jon Gjengset: <https://www.youtube.com/c/JonGjengset>
- Tris <https://www.youtube.com/@NoBoilerplate>



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1.1. Review test

1. Smart pointers

What is a smart pointer?

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What is a smart pointer?

A pointer with ownership.

Name 10 smart pointers in Rust.

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A pointer with ownership.

Name 10 smart pointers in Rust.

Box<T>, Rc<T>, Arc<T>, RefCell<T>, Mutex<T>, RwLock<T>, Weak<T>, Cow<T>, String, Vec<T>.

A pointer is generally speaking a type that implements Deref trait.

```
1 pub trait Deref {  
2     type Target: ?Sized;  
3     fn deref(&self) -> &Self::Target;  
4 }
```

(playground link)

Which types implement Deref but are NOT smart pointers?

1.1. Review test

What is a smart pointer?

A pointer with ownership.

Name 10 smart pointers in Rust.

Box<T>, Rc<T>, Arc<T>, RefCell<T>, Mutex<T>, RwLock<T>, Weak<T>, Cow<T>, String, Vec<T>.

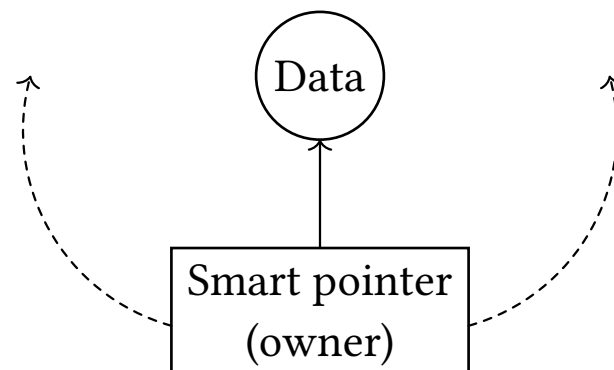
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```
1 pub trait Deref {  
2     type Target: ?Sized;  
3     fn deref(&self) -> &Self::Target;  
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```

(playground link)

Which types implement Deref but are NOT smart pointers?

References (&T, &mut T) are primitive pointers without ownership. Cell<T> and MaybeUninit<T> don't implement Deref.



1.3. Exercises

Solve the exercises in `session-4/examples/` (if you haven't already) in 30 minutes.

Are you done or bored? Implement your own smart pointer. It should own its data and provide shared access to it (single threaded):

- Use `dealloc` and `alloc` for unsafe memory management
- Use `AtomicUsize` for reference counting

Solution in `main.rs` (don't read before trying)

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2.1. Borrowing with functions

```
1 fn borrows(x: &i32) {  
2     dbg!(x);  
3 }  
4  
5 fn main() {  
6     let mut val = 123;  
7  
8     // Borrow `val` for the function call.  
9     borrows(&val);  
10  
11    // Borrow has ended and we're free to mutate.  
12    val += 5;  
13 }
```

(playground link)

```
1  fn identity(x: &i32) -> &i32 {  
2      x  
3  }  
4  
5  fn main() {  
6      let mut x = 123;  
7      let out = identity(&x);  
8      // x = 5; //   `x` is still borrowed!  
9      dbg!(out);  
10 }
```

(playground link)

How does the compiler check out is not a dangling reference?

```
1  fn identity(x: &i32) -> &i32 {  
2      x  
3  }  
4  
5  fn main() {  
6      let mut x = 123;  
7      let out = identity(&x);  
8      // x = 5; //   `x` is still borrowed!  
9      dbg!(out);  
10 }
```

(playground link)

How does the compiler check out is not a dangling reference?

By checking the lifetimes of the references.

Where are the lifetimes?

2.2. Returning Borrows

```
1  fn identity(x: &i32) -> &i32 {  
2      x  
3  }  
4  
5  fn main() {  
6      let mut x = 123;  
7      let out = identity(&x);  
8      // x = 5; //  `x` is still borrowed!  
9      dbg!(out);  
10 }
```

(playground link)

How does the compiler check out is not a dangling reference?

By checking the lifetimes of the references.

Where are the lifetimes?

They are hidden in the signature (**lifetime elision**).

2.3. Multiple Borrows

```
1 fn multiple(a: &i32, b: &i32) -> &i32 {
2     todo!("Return either `a` or `b`")
3 }
4
5 fn main() {
6     let mut a = 5; let mut b = 10;
7     let r = multiple(&a, &b);
8
9     // Which one is still borrowed?
10    // Should either mutation be allowed?
11    a += 7; b += 7;
12    dbg!(r);
13 }
```

(playground link)

Why does this code not compile?

2.3. Multiple Borrows

```
1  fn multiple(a: &i32, b: &i32) -> &i32 {  
2      todo!("Return either `a` or `b`")  
3  }  
4  
5  fn main() {  
6      let mut a = 5; let mut b = 10;  
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9      // Which one is still borrowed?  
10     // Should either mutation be allowed?  
11     a += 7; b += 7;  
12     dbg!(r);  
13 }
```

(playground link)

Why does this code not compile?

Compiler cannot derive which reference to return, so both are considered borrowed.

2.4. Borrow both

```
1 fn pick<'a>(c: bool, a: &'a i32, b: &'a i32) -> &'a i32 {
2     if c { a } else { b }
3 }
4
5 fn main() {
6     let mut a = 5; let mut b = 10;
7     let r = pick(true, &a, &b);
8     // Which one is still borrowed?
9     // Should either mutation be allowed?
10    // a += 7;
11    // b += 7;
12    dbg!(r);
13 }
```

(playground link)

Which argument is borrowed?

2.4. Borrow both

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9     // Should either mutation be allowed?  
10    // a += 7;  
11    // b += 7;  
12    dbg!(r);  
13 }
```

(playground link)

Which argument is borrowed?

Both a and b are borrowed for the lifetime of r. Compiler looks only at signature.



2.5. Borrow One

It is possible to borrow from only one argument by introducing a second lifetime. (Even why they are both references)

Open example file for demonstration: `find-nearest.rs`

```
1  fn only_args(a: &i32, b: &i32) {  
2      todo!();  
3  }  
4  
5  fn identity(a: &i32) -> &i32 {  
6      a  
7  }  
8  
9  struct Foo(i32);  
10 impl Foo {  
11     fn get(&self, other: &i32) -> &i32  
12     {  
13         &self.0  
14     }  
15 }
```

(playground link)

2.6. Lifetime Elision

```
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```

(playground link)

2.6.1. Rules for lifetime elision

... for references in argument or return position:

- Each argument which does not have a lifetime annotation is given one.
- If there is only **one argument lifetime**, it is given to all un-annotated return values.
- If there are **multiple argument lifetimes**, but the **first one is for self**, that lifetime is given to all un-annotated return values.

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Warning

Lifetime elision rules are tricky and very important! Learn them by hard!

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Warning

Lifetime elision rules are tricky and very important! Learn them by hard!

Complete elided lifetimes in the example code in example file `lifetime-elision.rs`.

2.7. Lifetimes in Data Structures

2. Lifetimes (new)

```
1  #[derive(Debug)]
2  enum HighlightColor {
3      Pink,
4      Yellow,
5  }
6
7  #[derive(Debug)]
8  struct Highlight<'document> {
9      slice: &'document str,
10     color: HighlightColor,
11 }
12
13 fn main() {
14     let doc = String::from("The quick brown fox jumps over the lazy dog.");
15     let noun = Highlight { slice: &doc[16..19], color: HighlightColor::Yellow };
16     let verb = Highlight { slice: &doc[20..25], color: HighlightColor::Pink };
17     // drop(doc);
18     dbg!(noun);
19     dbg!(verb);
20 }
```

[\(playground link\)](#)

What does the lifetime 'document represent?

2.7. Lifetimes in Data Structures

2. Lifetimes (new)

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1  #[derive(Debug)]
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13 fn main() {
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17     // drop(doc);
18     dbg!(noun);
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20 }
```

[\(playground link\)](#)

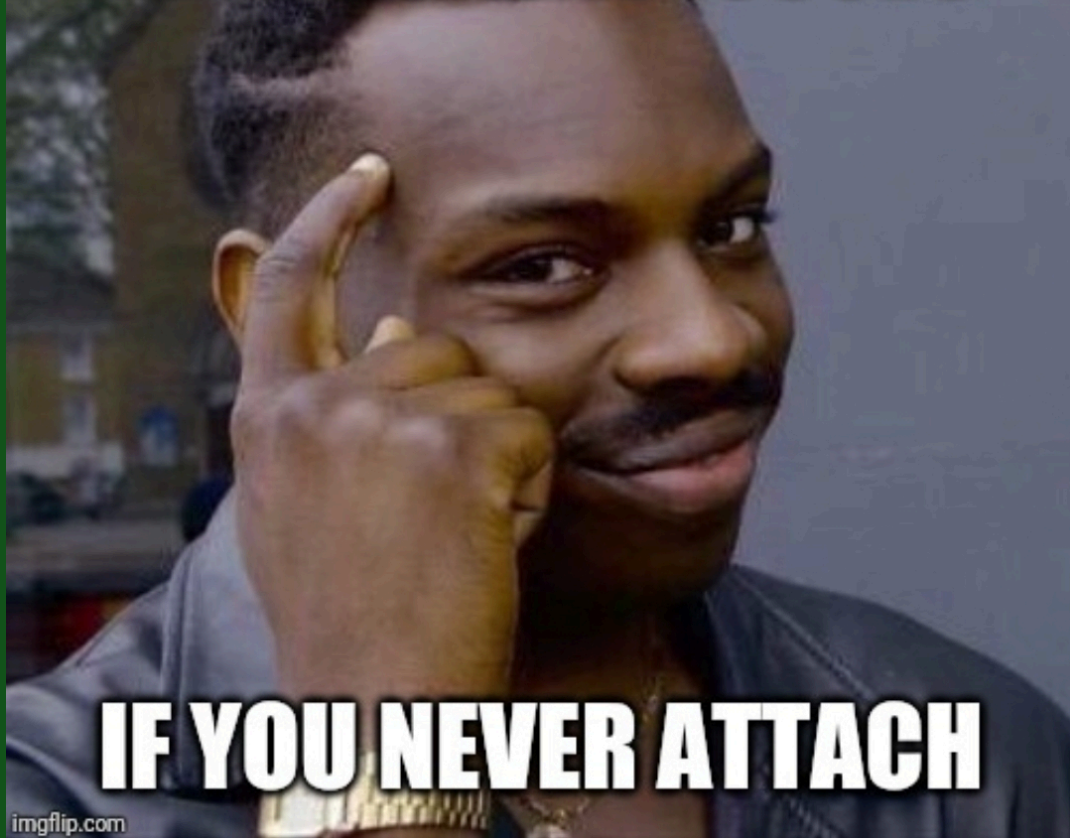
What does the lifetime 'document represent?

The lifetime of the slice referring to doc String in main() needs to exceed any instance of Highlight (that contains that slice).

2.8. Guidelines

- Types with borrowed data **force users to hold on to the original** data. This can be performant (but harder):
 - creating lightweight views
 - allocation-free parsers
- When possible, make data structures own their data directly.

**CAN'T HAVE
ATTACHMENT ISSUES**



IF YOU NEVER ATTACH

- introductory exercises in Rustlings chapter 16: Lifetimes (clone the Rustlings repo)
- advanced exercise in example file `protobuf-parsing.rs`. (in this session's examples folder)

2.10. More reading material

For those interested: read this Blog post by PretzelHamer titled “Common Rust Lifetime Misconceptions”

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Not necessarily. The compiler checks for safety, but does not know your intentions.

It is possible to write large Rust programs without ever using lifetimes. True or false?

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Not necessarily. The compiler checks for safety, but does not know your intentions.

It is possible to write large Rust programs without ever using lifetimes. True or false?

Not quite true. Lifetimes are almost always omitted because of lifetime elision rules. You need them for implementing iterators.

2.12. Demonstration

```
1  struct ByteIter<'a> {
2      remainder: &'a [u8]
3  }
4
5  impl<'a> ByteIter<'a> {
6      fn next(&mut self) -> Option<&u8> {
7          if self.remainder.is_empty() {
8              None
9          } else {
10             let byte = &self.remainder[0];
11             self.remainder = &self.remainder[1..];
12             Some(byte)
13         }
14     }
15 }
16
17 fn main() {
18     let mut bytes = ByteIter { remainder: b"1" };
19     assert_eq!(Some(&b'1'), bytes.next());
20     assert_eq!(None, bytes.next());
21 }
```

[\(playground link\)](#)

2.12. Demonstration

```
1 fn main() {  
2     let mut bytes = ByteIter { remainder: b"1123" };  
3     let byte_1 = bytes.next();  
4     let byte_2 = bytes.next();  
5     if byte_1 == byte_2  
6         // do something  
7 }  
8 }
```

(playground link)

Will it compile?

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5     if byte_1 == byte_2
6         // do something
7     }
8 }

```

(playground link)

Will it compile?

No.

```

1 error[E0499]: cannot borrow `bytes` as mutable more than once at a time
2   --> src/main.rs:20:18
3   |
4 19 |     let byte_1 = bytes.next();
5   |                  ----- first mutable borrow occurs here
6 20 |     let byte_2 = bytes.next();
7   |                  ^^^^^^ second mutable borrow occurs here
8 21 |     if byte_1 == byte_2 {
9   |         ----- first borrow later used here

```

(playground link)

2.12. Demonstration

Fill in missing lifetimes and name them properly

```
1 struct ByteIter<'remainder> {  
2     remainder: &'remainder [u8]  
3 }  
4  
5 impl<'remainder> ByteIter<'remainder> {  
6     fn next<'mut_self>(&'mut_self mut self) -> Option<'mut_self u8> {  
7         if self.remainder.is_empty() {  
8             None  
9         } else {  
10             let byte = &self.remainder[0];  
11             self.remainder = &self.remainder[1..];  
12             Some(byte)  
13         }  
14     }  
15 }
```

(playground link)

```
1 struct Wrapper<T> {  
2     value: T,  
3 }
```

(playground link)

Wherever the type generic `T` appears only owned types may be used. True or false?

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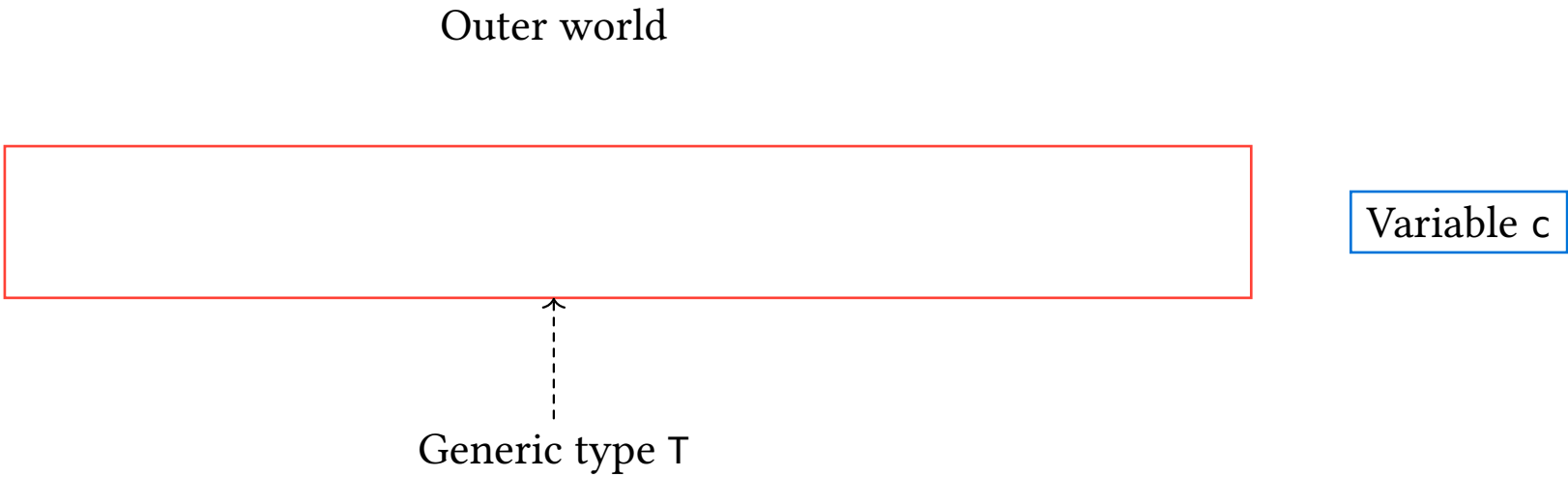
False. `T` may contain references too.


```
1 struct Wrapper<T> {  
2     value: T,  
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```

(playground link)

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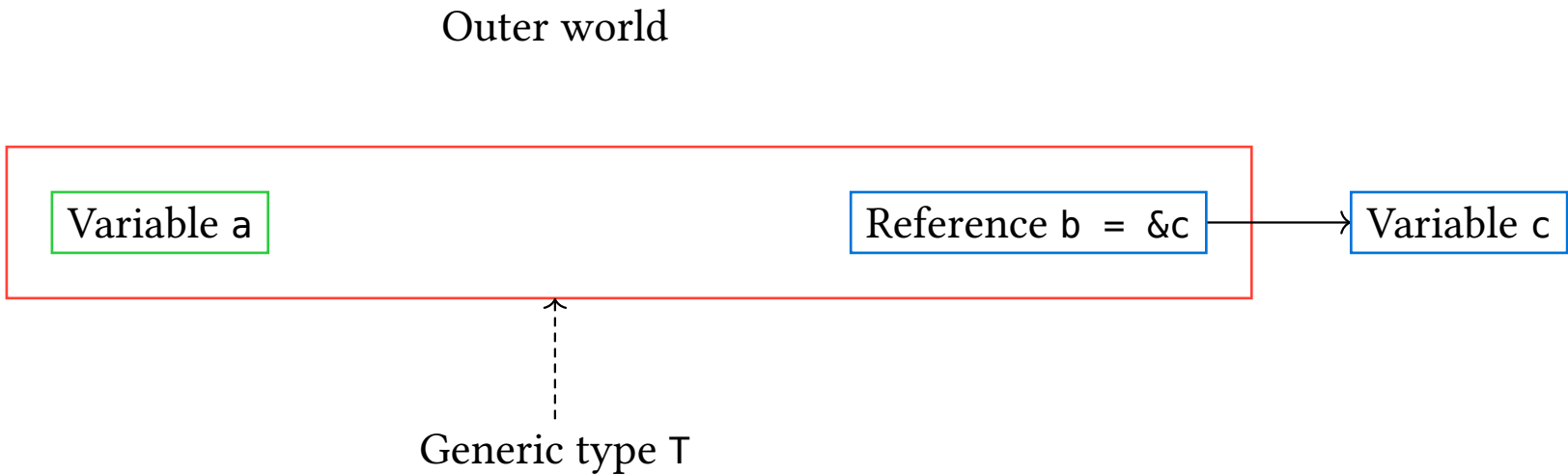


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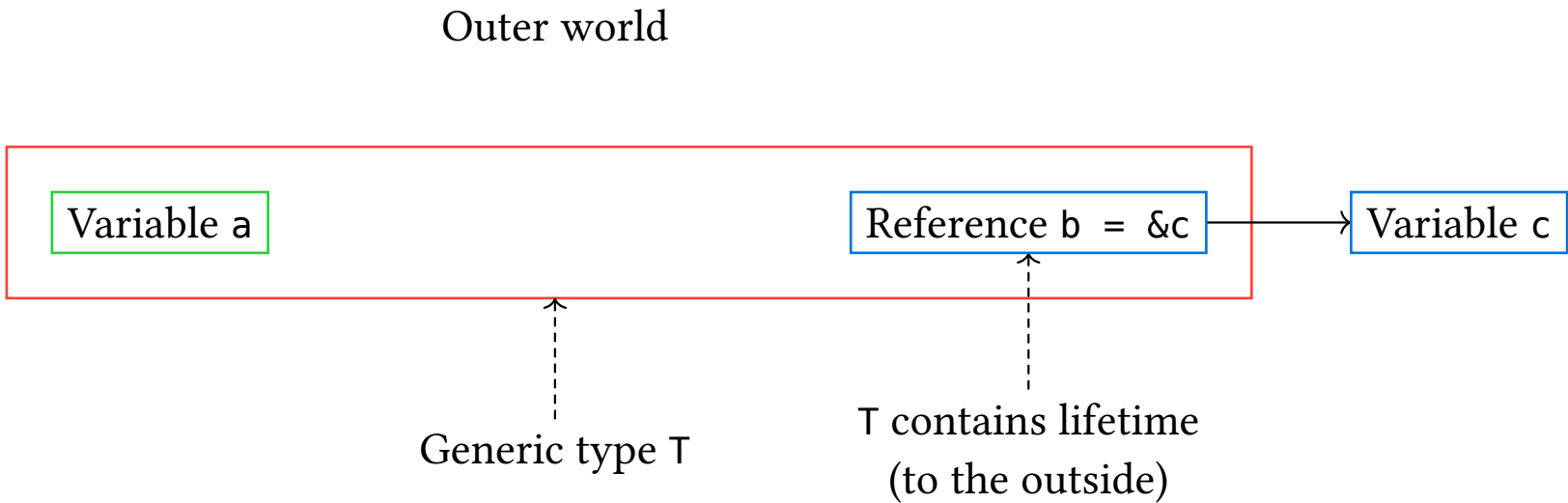


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(playground link)

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In this situation: T: 'out where 'out is the lifetime of b (may be implicit in T)

Type compatibility for generic type parameters:

Type Variable	T	&T	&mut T
Examples	i32, &i32, &mut i32, &&i32, ...	&i32, &&i32, &&mut i32, ...	&mut i32, &mut &mut i32, &mut &i32, ...

```
1 trait Trait {}
2 impl<T> Trait for T {}
3 impl<T> Trait for &T {} // ✗
4 impl<T> Trait for &mut T {} // ✗
```

(playground link)

Why do the last two impls not compile?

Type compatibility for generic type parameters:

Type Variable	T	&T	&mut T
Examples	i32, &i32, &mut i32, &&i32, ...	&i32, &&i32, &&mut i32, ...	&mut i32, &mut &mut i32, &mut &i32, ...

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3 impl<T> Trait for &T {} // ✗
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```

(playground link)

Why do the last two impls not compile?

The compiler doesn't allow us to define an implementation of Trait for &T and &mut T since it would conflict with the implementation of Trait for T which already includes all of &T and &mut T

2.14. 'static lifetime

The 'static is a special lifetime (not to be confused with the 'static keyword for variables). It stands for the longest / maximum lifetime.

If T : 'static then T must be valid for the entire program. True or false?

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Examples: primitive types: `i32`, `f64`, `bool`, `char`.

Conclusion:

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Examples: primitive types: `i32`, `f64`, `bool`, `char`.

Conclusion:

- `T: 'static` should be read as *"T can live at least as long as a 'static lifetime"*
- if `T: 'static` then `T` can be a borrowed type with a 'static lifetime *or* an owned type
- since `T: 'static` includes owned types that means `T`
 - can be dynamically allocated at run-time
 - does not have to be valid for the entire program
 - can be safely and freely mutated
 - can be dynamically dropped at run-time
 - can have lifetimes of different durations



2.15. Recent change in compiler

Since 2025 (Rust edition 2024), `impl` blocks in the return type position now capture lifetimes in argument position automatically.

```
1 fn multiply_adapter(  
2     iter: impl Iterator<Item = i32>,  
3     factor: &Wrapper,  
4 ) -> impl Iterator<Item = i32> {  
5     iter.map(move |x| x * factor.factor)  
6 }
```

(playground link)

See example `impl-return.rs` for demonstration.

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3.1. Motivation

If you want to iterate over the contents of an array, you'll need to define:

- Some state to keep track of where you are in the iteration process, e.g. an index.
- A condition to determine when iteration is done.
- Logic for updating the state of iteration each loop.
- Logic for fetching each element using that iteration state.

```
1 for (int i = 0; i < array_len; i += 1) {  
2     int elem = array[i];  
3 }
```

(playground link)

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What is the closest C equivalent of this to Rust?

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```
1 for (int i = 0; i < array_len; i += 1) {  
2     int elem = array[i];  
3 }
```

(playground link)

What is the closest C equivalent of this to Rust?

It uses a pointer that is incremented instead of an index.

```
1 for (int *ptr = array; ptr < array + len; ptr += 1) {  
2     int elem = *ptr;  
3 }
```

(playground link)

3.2. Iterator trait

```
1 struct SliceIter<'s> {
2     slice: &'s [i32],
3     i: usize,
4 }
5
6 impl<'s> Iterator for SliceIter<'s> {
7     type Item = &'s i32;
8
9     fn next(&mut self) -> Option<Self::Item> {
10         if self.i == self.slice.len() {
11             None
12         } else {
13             let next = &self.slice[self.i];
14             self.i += 1;
15             Some(next)
16         }
17     }
18 }
19 (playground link)
```

The Iterator trait defines how an object can be used to **produce a sequence of values**.

```
1 fn main() {
2     let slice = &[2, 4, 6, 8];
3     let iter = SliceIter { slice, i:
4         0 };
5     for elem in iter {
6         dbg!(elem);
7     }
8 } (playground link)
```

Why are Rust iterators lazy?

3.2. Iterator trait

Why are Rust iterators lazy?

You can call `next()` to get the next element only when you need it.

What happens when you call `next()` after the iterator returned `None`

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Why are Rust iterators lazy?

You can call `next()` to get the next element only when you need it.

What happens when you call `next()` after the iterator returned `None`

We don't know. Only fused iterators guarantee `None` forever after.

```
1 fn main() {  
2     let mut iter = SliceIter { slice: &[1, 2], i: 0 };  
3     assert_eq!(Some(&1), iter.next());  
4     assert_eq!(Some(&2), iter.next());  
5     assert_eq!(None, iter.next());  
6     assert_eq!(???, iter.next()); // What happens here?  
7 }
```

(playground link)

Give an example of an infinite iterator.

3.3. Generators

Give an example of an infinite iterator.

The range `0..` is an infinite iterator of integers starting from 0.

Iterators can also be generated with coroutines (called generators in Rust):

```
1 fn counter() -> impl Iterator<Item = i32> {  
2     gen {  
3         let mut count = 0;  
4         loop {  
5             yield count;  
6             count += 1;  
7         }  
8     }  
9 }
```

(playground link)

(Also an infinite iterator)

3.4. Helpers

the Iterator trait provides **helper methods** that can be used to transform iterators:

```
1 fn main() {  
2     let result: i32 = (1..=10) // Create a range from 1 to 10  
3     .filter(|x| x % 2 == 0) // Keep only even numbers  
4     .map(|x| x * x) // Square each number  
5     .sum(); // Sum up all the squared numbers  
6  
7     println!("The sum of squares of even numbers from 1 to 10 is: {}", result);  
8 }                                     (playground link)
```

What is another name for iterator helpers?

3.4. Helpers

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8 } (playground link)
```

What is another name for iterator helpers?

Adapters (or operators or combinators).

How many adapters are there in the standard library?

3.4. Helpers

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8 }                                     (playground link)
```

What is another name for iterator helpers?

Adapters (or operators or combinators).

How many adapters are there in the standard library?

Approximately 40.

3.5. Collect

Build collections from iterators:

```
1 fn main() {  
2     let primes = vec![2, 3, 5, 7];  
3     let prime_squares = primes.into_iter().map(|p| p * p).collect::<Vec<_>>();  
4     println!("prime_squares: {prime_squares:?}");  
5 }
```

(playground link)

What are the collections that we can collect into?

3.5. Collect

Build collections from iterators:

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(playground link)

What are the collections that we can collect into?

Vectors, HashMaps, HashSets, BtreeMap, BtreeSet, VecDeque, ... (your own collections too!)

Target collection type is either implicit or specified with:

3.5. Collect

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(playground link)

What are the collections that we can collect into?

Vectors, HashMaps, HashSets, BtreeMap, BtreeSet, VecDeque, ... (your own collections too!)

Target collection type is either implicit or specified with:

- Turbofishing: `collect::<Vec<_>>()`
- Type annotation: `let v: Vec<_> = iterator.collect();`

3.6. Consuming adapters

Consume the iterator and produce a final value:

```
1 let nums = vec![1, 2, 3, 4, 5];
2 nums.iter().collect::<Vec<_>>(); // collect into collection
3 nums.iter().sum::<i32>();         // reduce to single value
4 nums.iter().count();              // count elements
5 nums.iter().fold(0, |acc, x| acc + x); // general reduction
6 nums.iter().for_each(|x| println!("{}", x)); // side effects      (playground link)
```

Other examples: max, min, find, any, all, partition, reduce.

Why are they called “consuming”?

3.6. Consuming adapters

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```

Other examples: max, min, find, any, all, partition, reduce.

Why are they called “consuming”?

They call `next()` until `None`.

3.7. Nonconsuming adapters

Transform an iterator into another iterator (lazy):

```
1 let nums = vec![1, 2, 3, 4, 5];
2 nums.iter().map(|x| x * 2);           // transform elements
3 nums.iter().filter(|x| *x % 2 == 0); // keep matching elements
4 nums.iter().take(3);                  // limit to first N
5 nums.iter().skip(2);                  // skip first N
6 nums.iter().enumerate();              // add indices
7 nums.iter().chain([6, 7].iter());     // concatenate
8 nums.iter().zip(['a', 'b'].iter());   // pair up                (playground link)
```

Other examples: flatten, flat_map, peekable, rev, cycle, cloned.

Why are they called “nonconsuming”?

3.7. Nonconsuming adapters

Transform an iterator into another iterator (lazy):

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1 let nums = vec![1, 2, 3, 4, 5];  
2 nums.iter().map(|x| x * 2);           // transform elements  
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8 nums.iter().zip(['a', 'b'].iter());   // pair up
```

(playground link)

Other examples: flatten, flat_map, peekable, rev, cycle, cloned.

Why are they called “nonconsuming”?

They don't run until a consuming adapter is called.

What is the cost of using lots of iterator adapters?

3.8. Overhead

What is the cost of using lots of iterator adapters?

There is no runtime overhead because most adapters work by reference and can be inlined or optimized by the compiler.

Exercises: Rustlings, chapter 18: Iterators

3.9. AsyncIterator adapters

Fragment from my EuroRust 2025 talk on functional async programming in Rust (github.com/wvhulle/streams-eurorust-2025/):

```
1 let results: Vec<String> = tcp_stream
2   .filter_map(|conn| ready(conn.ok()))
3   .filter(|stream| ready(should_process(stream)))
4   .then(|stream| process_stream(stream))
5   .filter_map(|result| ready(result.ok()))
6   .filter(|msg| ready(msg.len() > 10))
7   .take(5)
8   .collect()
9   .await;                                     (playground link)
```

Benefits:

- Each operation is isolated
- Testable
- Reusable

3.10. IntoIterator

IntoIterator defines how to create an iterator for a type:

```
1  struct Grid {  
2      x_coords: Vec<u32>,  
3      y_coords: Vec<u32>,  
4  }  
5  
6  impl IntoIterator for Grid {  
7      type Item = (u32, u32);  
8      type IntoIter = GridIter;  
9      fn into_iter(self) -> GridIter {  
10         GridIter { grid: self, i: 0, j: 0 }  
11     }  
12 }
```

(playground link)

3.11. The “actual” / physical iterator

```
1  struct GridIter {
2      grid: Grid,
3      i: usize,
4      j: usize,
5  }
6
7  impl Iterator for GridIter {
8      type Item = (u32, u32);
9
10     fn next(&mut self) -> Option<(u32, u32)> {
11         if self.i >= self.grid.x_coords.len() {
12             self.i = 0;
13             self.j += 1;
14             if self.j >= self.grid.y_coords.len() {
15                 return None;
16             }
17         }
18         let res = Some((self.grid.x_coords[self.i], self.grid.y_coords[self.j]));
19         self.i += 1;
20         res
21     }
22 }
```

(playground link)

3.12. Different IntoIterator implementations

Iterating over a vector like this will consume it:

```
1 let v = vec![1, 2, 3];  
2 for elem in v {  
3     dbg!(elem);  
4 }  
5 // v is no longer usable here (playground link)
```

Iterating over a reference to a vector will not consume it:

```
1 let v = vec![1, 2, 3];  
2 for elem in &v {  
3     dbg!(elem);  
4 }  
5 // v is still usable here (playground link)
```

3.13. Iterate over Grid by reference

You might have multiple ways to create an iterator for a type.

Usually, by value (consuming) or by reference (non-consuming):

```
1 impl<'a> IntoIterator for &'a Grid {  
2     type Item = (u32, u32);  
3     type IntoIter = GridRefIter<'a>;  
4     fn into_iter(self) -> GridRefIter<'a> {  
5         GridRefIter { grid: self, i: 0, j: 0 }  
6     }  
7 }
```

(playground link)

From a mutable reference (non-consuming + mutable):

```
1 impl<'a> IntoIterator for &'a mut Grid {  
2     type Item = (u32, u32);  
3     type IntoIter = GridMutRefIter<'a>;  
4     fn into_iter(self) -> GridMutRefIter<'a> {  
5         GridMutRefIter { grid: self, i: 0, j: 0 }  
6     }  
7 }
```

(playground link)

See example file `into-iterator-ref.rs`.

3.14. Exercises

Solve the exercise in `tests/iterator-method-chaining.rs`

Create your own iterator by implementing `Iterator` and `IntoIterator` in `examples/custom-iterator.rs`

Build your own `Iterator` adapters in `examples/iterator-adapters.rs`